Su-Din Technology

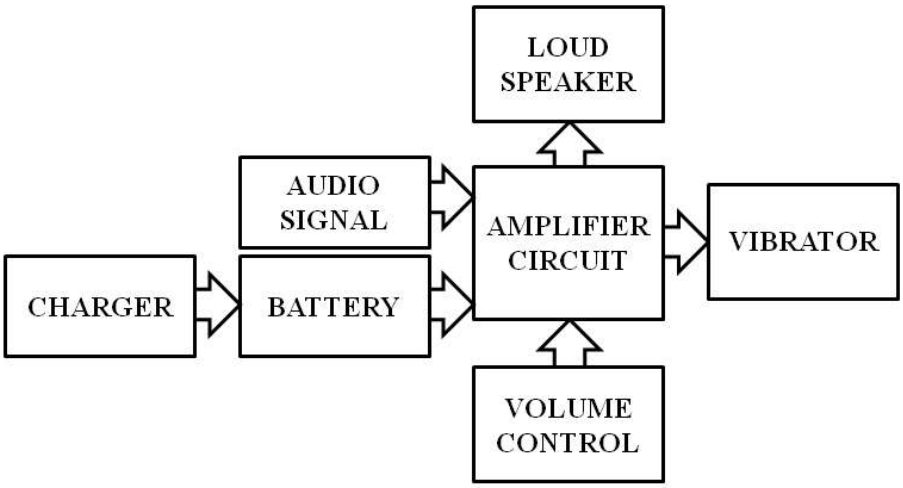
TEACHING SYSTEM FOR A DEAF STUDENT.

1 February, 2023 Anil 0 Comments 2 categories

Introduction

In this project by using engineering and modern technology, we are trying to build a hearing system for deaf and semi deaf people. This system is useful to teach a deaf student by providing recorded audio classes. Video classes can be enjoyed also. It is really a hundred percent practical entertainment system for enjoying music by teeth. For this our primary requirement will be a low power audio signal amplifier system. By taking the help of physics, understanding biological facts behind the hearing process and using electronics knowledge let's make the project successful. The components used here are easily available and cheap.

Block diagram



Components required

Capacitor 10 µF, 220 µF, 0.22µF

LM386

Resistors 470 Ω, 1.2 kΩ, 50 Ω

DC Motor used as a vibrator

Variable resistor 10 kΩ

LED

Switch

Loud speaker 4 Ω, 2.5 inch

Audio pin (stereo)

3.7 volt Li-ion battery charger.

Li-ion battery

Components Description

LM 386

The op-amp used here is LM386 which is a general purpose op-amp and easily available in market. It is manufactured by Texas

Instruments. It is a low voltage audio power amplifier. It can operate with a supply voltage from 5V to 18 V. The gain is internally set to 20 to keep external part count low, but the addition of an external resistor and capacitor between pins 1 and 8 will increase the gain to any value from 20 to 200. It is generally used in AM-FM radio amplifiers.

Resistor

A resistor restricts the flow of current. The resistors used here are carbon film and quarter watt type.

Variable resistor

The resistance value can be adjusted by rotating the knob. It is used here as a volume control.

LED (Light Emitting Diode)

It is a transducer which converts electrical energy into light energy.

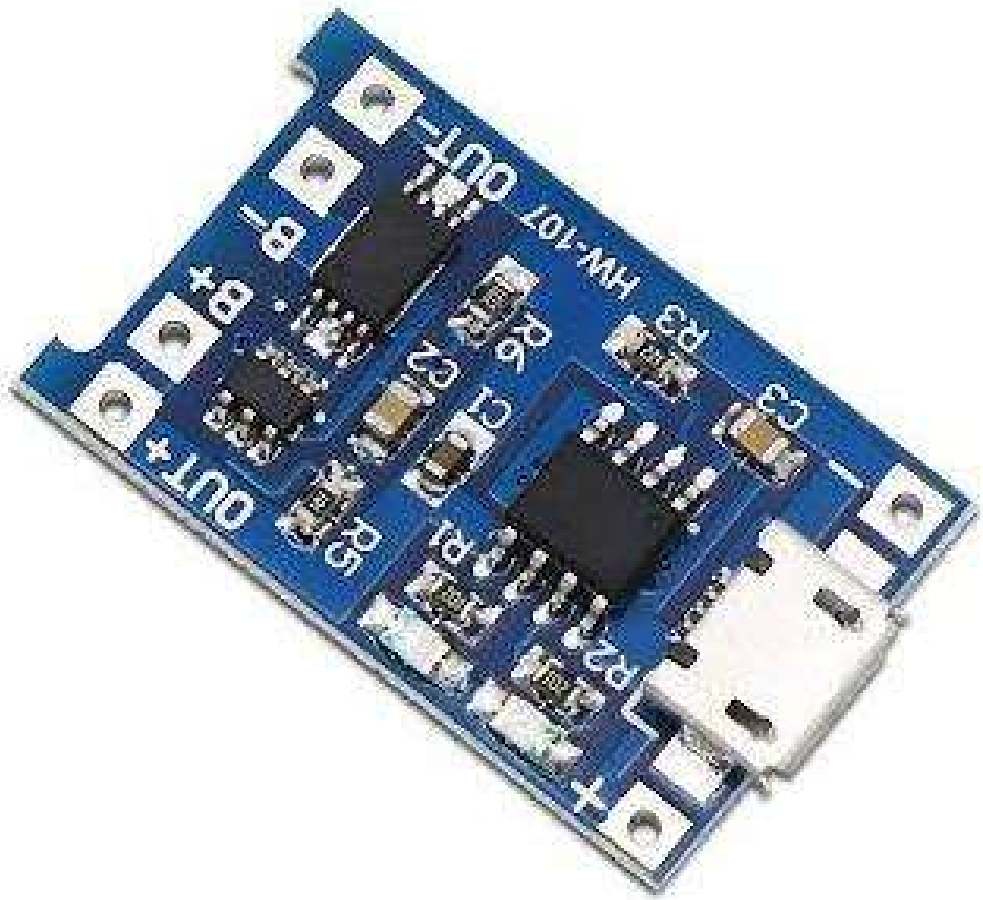
Capacitor

A capacitor stores electric charge. Capacitors 10 μF, 220 μF are electrolytic type but 0.22μF is a ceramic capacitor.

DC Motor used as vibrator

A motor is a device which converts electrical power into mechanical power. Here we have used a small 3.7 Volt permanent magnet type DC motor. The armature of the motor vibrates according to the music signal.

3.7 volt Li-ion battery charger



This charger is used for charging the Li-ion battery.
We have used a B-type mobile charger as input.

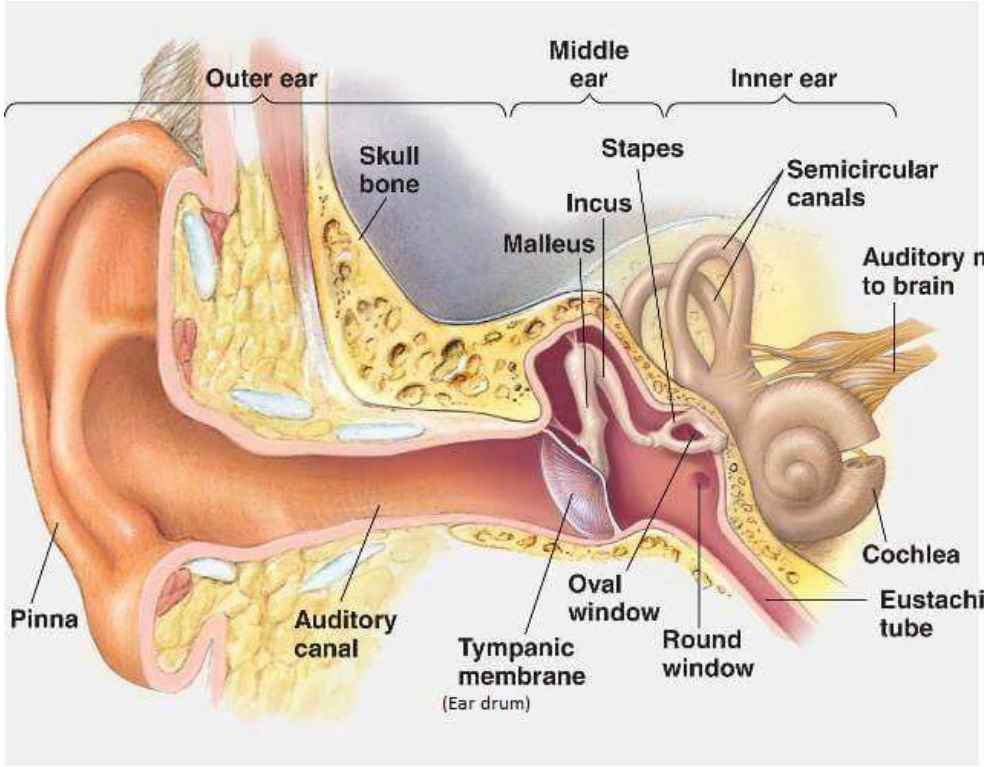
Li-ion battery

A battery stores electric charge. When we discharge a battery, the stored chemical energy is converted into electrical energy. When we charge a battery, electrical energy is converted into chemical energy. A battery always supplies us DC. Here we have used a 3.7 Volt Li-ion battery.

Loud speaker

Here we have used a 2.5 inch speaker. A speaker converts electrical signal into audible sound. The main source of sound is vibration. When the diaphragm of speaker vibrates, it produces longitudinal waves which propagate through air medium.

The biology behind deafness.

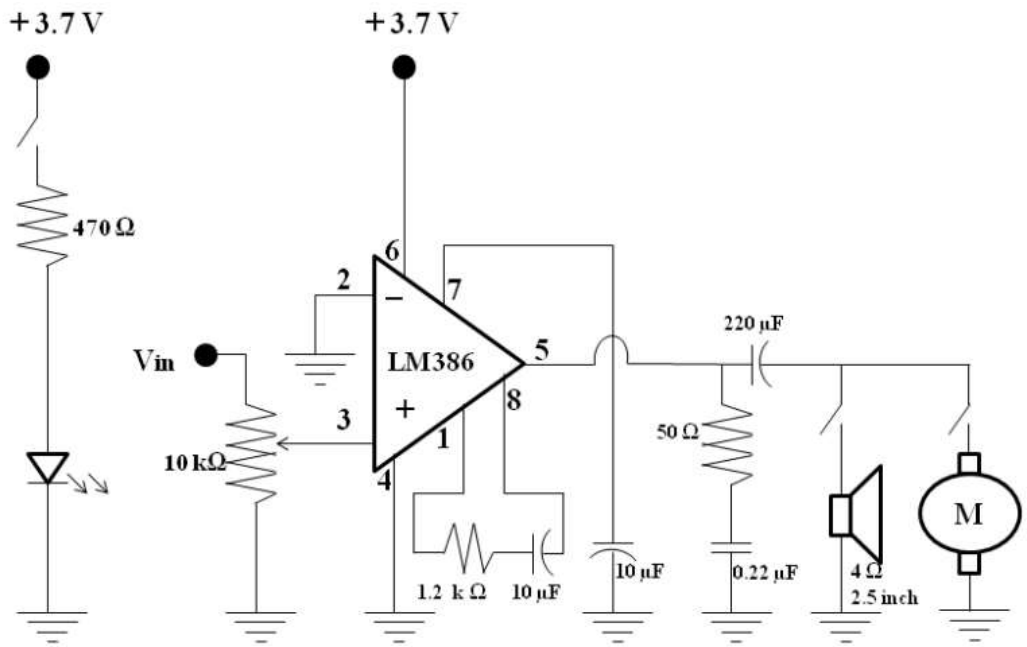


When the middle ear is not functioning correctly, sound wave cannot travel through the auditory (hearing) nerve. This is called conductive hearing loss. But this type of hearing loss is usually temporary. The hearing generally comes to normal state with medical treatment. If middle ear doesn't function well due to injury or shock or infection, one person becomes deaf. In that case if the inner ear is at good condition then sound can be conducted through skull by teeth to the inner ear. The brain can feel the sound if the working cochlea receives the vibration through the skull. In this way a deaf person can enjoy the music through vibration in teeth followed by skull in a non-conventional way.

The physics behind sound

The main source of sound is vibration. Everyone knows that the range of audio signal is 20 Hz to 20 kHz. A sound wave is an air pressure disturbance that originates from vibration. The vibration can be produced from a tuning fork, a guitar string, the column of air in an organ pipe, the head (or rim) of a snare drum, steam escaping from a radiator, the reed on a clarinet, the diaphragm of a loudspeaker, the vocal cords, or virtually anything that vibrates in a frequency range that is audible to a listener (roughly 20 to 20,000 cycles per second for humans). The two conditions that are required for the generation of a sound wave are a vibratory disturbance and an elastic medium. The most familiar elastic medium is air.

Working of the circuit.



The components are mounted on a printed circuit board. When an amplified music signal is given to a dc motor, the motor vibrates accordingly. This vibration is captured by teeth and conducted through skull to the cochlea. The music signal can be regulated through the regulator. The intensity of vibration of the dc motor is controlled by the variable resistor. So, the volume of music is increased or decreased accordingly by adjusting the 10 kΩ variable resistor. Hear the LM 386 IC is the heart of the amplifier circuit. The input audio signal is provided through a cable with a stereo type audio pin. The music source may be a mobile phone or an ipod mp3 player.

The amplifier circuit operates through 3.7 V DC supply. The charger used here converts 5 V DC from mobile charger into 4.2 V DC and feeds

the battery. The loud speaker used here is for experimental purpose only. That's why both the loud speaker and motor are provided with series switches. Speaker is operated to confirm that the music system is functioning or not. When the motor functions the loud speaker must be switched off.

How an op-amp works

An operational amplifier is an integrated circuit. Its primary work is to amplify weak electric signals. An operational amplifier consists of two input pins and one output pin. Its basic role is to amplify and output the voltage difference between the two input pins. The op-amp has 2 input, the +ve input, and the -ve input. If the +ve input's voltage is higher than the -ve input's voltage, the output will be high. Even an op-amp can detect a voltage difference of 0.0001 volts.

Advantages

1. It is a music system for deaf and semi deaf people.
2. Due to its compact size, it is portable and useful to teach a deaf student by providing recorded audio classes.
3. Video classes(YouTube Learning) can be enjoyed also.

Contact:

Er. Anil Kumar Prusty

Phone: +91 **9861004895**

Tags: health and innovation | Inspire award project | sudin technology

Category: Do it yourself, Science and Technology

One Reply to "Teaching System For A Deaf Student."

Anil

FEBRUARY 1, 2023 AT 11:22 AM

This compact size hearing aid is useful for deaf students.

Reply

Leave a Reply

Your email address will not be published. Required fields are marked *

Comment *

Name *

Email *

Website

Save my name, email, and website in this browser for the next time I comment.

Post Comment

Previous: Solar Panel Based Electric Tricycle for Physically Challenged People

Next: ବୃଦ୍ଧାମାନଙ୍କ ପାଇଁ ଚାଲିବା ଠେଙ୍ଗା

Related Posts

Face Mask Disinfection

Introduction In this project we are giving dry heat to…

Read More

Automatic Flood Resistant House.

Introduction : Considering the harmful effects of flood, we have…

Read More

ଏକ୍ସାମ ପ୍ୟାଡ ଫ୍ରମ୍ କର୍ନ କବ୍

Designer:Er. Anil Kumar PrustyWhatsApp: +91 9861004895 https://www.youtube.com/shorts/y8V_WIkB8c8?feature=share

Read More

Search …

Search

RECENT POSTS

- ସ୍ମାର୍ଟ ଭିଲେଜ (Smart Village)

- ଗଛ ଓପଡ଼ା ଯନ୍ତ୍ର (Tree eradicating gadget)
- ବର୍ଷା ଓ ଉଚ ତାପମାତ୍ରା ରୁ ଚାରା ଗଛର ସୁରକ୍ଷା (Protection of seedlings from rain and high temperature)
- କାଟେଲ କାଉଣ୍ଟିଙ୍ଗ ମେସିନ (ପଶୁ ଗଣନ ଯନ୍ତ୍ର)
- ଅଟୋମେଟିକ କ୍ଲିନଡ଼ ଟଏଲେଟ

RECENT COMMENTS

- 9gag downloader on Bound to clean toilet
- Anil on ବୃଦ୍ଧାମାନଙ୍କ ପାଇଁ ଚାଲିବା ଠେଙ୍ଗା
- Anil on Teaching System For A Deaf Student.
- Anil on Solar Panel Based Electric Tricycle for Physically Challenged People
- Anil on Automatic Garbage Disposal System.

ARCHIVES

- February 2026
- November 2025
- August 2025

- July 2025
- May 2025
- February 2025
- October 2024
- September 2024
- August 2024
- July 2024
- May 2024
- March 2024
- January 2024
- December 2023
- October 2023
- September 2023
- August 2023
- July 2023
- June 2023
- April 2023
- March 2023
- February 2023

- November 2022
- October 2022
- September 2022
- August 2022
- July 2022
- June 2022
- May 2022
- August 2021
- June 2021
- May 2021

CATEGORIES

- Do it yourself
- Electrical Engineering
- INSPIRE AWARD PROJECT
- Interaction with students
- Plantation program
- Power Electronics

- Research and Development
- Robotics
- Science and Technology
- Science Project
- Uncategorized
- ଓଡିଆ ବିଜ୍ଞାନ ଲେଖା

META

- Log in
- Entries feed
- Comments feed
- WordPress.org
- Online store powered by Ecwid

Search

Search …

Search

Archives

February 2026

November 2025

August 2025

July 2025

May 2025

February 2025

October 2024

September 2024

August 2024

July 2024

May 2024

March 2024

January 2024

December 2023

October 2023

September 2023

August 2023

July 2023

June 2023

April 2023

March 2023

February 2023

November 2022

October 2022

September 2022

August 2022

July 2022

June 2022

May 2022

August 2021

June 2021

May 2021

Meta

Log in

Categories

Do it yourself

Electrical Engineering

INSPIRE AWARD PROJECT

Interaction with students

Plantation program

Power Electronics

Research and Development

Robotics

Science and Technology

Science Project

Uncategorized

ଓଡିଆ ବିଜ୍ଞାନ ଲେଖା

Educational Institute WordPress Theme By Themespride